

Disentangling Representation Evolution in Transformers through Directional Decomposition

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Abstract

Transformer layers evolve representations through learned additive transformations that combine parallel scaling and perpendicular steering. Across pretrained models, we find that learned updates consistently contain substantial parallel components. To test whether these components are redundant or behaviorally important, we decompose updates in two complementary spaces: residual space, analyzing complete sub-layer updates relative to the incoming hidden state; and attention value space, analyzing pre-output-projection aggregation relative to the token’s own value. Targeted edits reveal a consistent directional and space-dependent asymmetry: perpendicular edits are consistently disruptive, whereas parallel edits are comparatively redundant, especially in value space when preserving the direct self message while scaling cross-token aggregation. The same geometry informs compression and training: better-performing compressed models show lower perpendicular transformation error, and suppressing parallel attention components during from-scratch pretraining improves downstream performance across evaluated model scales, with the value-space variant strongest. Overall, this geometry connects update direction to editing robustness, compression diagnostics, and training-time intervention. Code is available in the [project repository](#).

1 Introduction

Transformer layers evolve representations through learned additive updates (Vaswani et al., 2017; Elhage et al., 2021) on top of residual streams (He et al., 2016). Geometrically, each update performs two distinct roles relative to the incoming representation: a *parallel component* that modulates magnitude while preserving direction, and a *perpendicular component* that redirects it into new

semantic subspaces. This dichotomy reframes representation evolution as a problem of *functional geometry*: decomposing learned updates by their directional role and testing their behavioral sensitivity through targeted interventions.

Across diverse pretrained models, our measurements reveal that learned updates consistently contain substantial components parallel to their incoming states. Geometrically, parallel updates amount to simple scalar rescaling, which the residual stream already provides at zero parameter cost (He et al., 2016; Elhage et al., 2021). Yet, heavily parameterized attention and MLP sub-layers actively allocate capacity to produce such components. This paradox prompts a fundamental question: are these parallel updates behaviorally essential to Transformer capabilities, or are they functionally redundant?

To resolve this question, we formulate directional decomposition across two complementary representation spaces: *residual space*, which analyzes sub-layer updates relative to the incoming hidden state, and *attention value space*, which analyzes internal value aggregation relative to the token’s own value representation. Using component-scaling interventions, we systematically modulate the parallel and perpendicular components in these spaces to measure their behavioral sensitivity.

Our interventions reveal a pronounced directional asymmetry across both representation spaces. Perpendicular scaling is consistently disruptive, confirming that orthogonal steering is essential to model capabilities. In contrast, parallel scaling is comparatively benign, leaving performance near baseline across a wide scaling interval. Within this parallel resilience, the degree of stability depends on the representation space: in attention value space, scaling the parallel component of cross-token aggregation keeps model behavior re-

*Work done while Shwai He was an intern at ByteDance.

markably close to baseline, whereas residual-space parallel scaling is comparatively less stable. Thus, parallel updates exhibit broad resilience compared to perpendicular steering, especially within internal value aggregation.

Finally, this directional geometry extends beyond inference-time editing to post-training compression and from-scratch pretraining. For model compression (Frantar et al., 2023; Lin et al., 2024; Sun et al., 2024; Men et al., 2024; He et al., 2026), decomposing update distortion reveals that better-performing quantized and pruned models consistently exhibit lower perpendicular error, whereas parallel error is far less discriminative. Motivated by this directional geometry, we evaluate parallel attention suppression during pretraining. Across model scales from 296M to 2.7B parameters, the value-space variant consistently yields lower validation-loss trajectories, and both variants improve downstream performance, with the value-space variant producing the largest gains.

In summary, the contributions of this work are as follows:

- This work reveals substantial direction-preserving components in learned transformer updates and maps their behavioral robustness across representation spaces.
- For attention updates, value-space decomposition separates direct self-value flow from cross-token aggregation, revealing a marked stability gap between value-space and residual-space edits.
- For compression and pretraining, this directional geometry extends beyond editing: perpendicular error tracks compressed-model quality, while parallel suppression consistently improves validation loss and downstream performance across model scales.

2 Related Work

Residual Stream Magnitude Scaling. Residual connections preserve input representations while allowing each block to add a learned transformation (He et al., 2016). In transformers, this creates a residual stream that carries information across layers and serves as the shared workspace for attention and feed-forward updates (Elhage et al., 2021). This preservation path is also a primary target for architectural scaling: mechanisms such as ReZero (Bachlechner et al., 2021), Layer-Scale (Touvron et al., 2021), and deep-transformer

normalization schemes (Xiong et al., 2020; Wang et al., 2024) adjust branch scale to stabilize optimization, while residual paths mitigate representational degeneracy such as rank collapse (Dong et al., 2021). These works establish the importance of residual pathways and branch magnitude for training stability. In contrast, we study a complementary directional geometric property of the learned update itself: whether it reinforces the incoming representation direction or steers it into orthogonal semantic subspaces.

Self-Directed Attention Value Flow. Attention explicitly aggregates information across tokens (Vaswani et al., 2017), but attention mass can also concentrate on special or persistent positions, as shown by attention-sink behavior in long-context inference (Xiao et al., 2024). Gated attention variants mitigate such sink behavior by modulating attention outputs (Qiu et al., 2025). However, even when attention sinks are reduced, large diagonal attention weights can still route substantial mass back to the current token itself. Exclusive Self-Attention (XSA) directly addresses this self-directed pathway by modifying value flow aligned with the current token (Zhai, 2026). In this work, we analyze this self-directed pathway from a directional geometric perspective, disentangling direction-preserving updates from direction-changing information.

3 Background

3.1 Parallel and Perpendicular Decomposition

Given a reference vector $\mathbf{z} \in \mathbb{R}^d$ and an update vector $\Delta \in \mathbb{R}^d$ within the same representation space, Δ admits a unique orthogonal decomposition into components parallel and perpendicular to $\mathbf{z}$:

$$\Delta_{\parallel} = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2} \mathbf{z} = \alpha \mathbf{z}, \quad \alpha = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2}, \quad (1)$$

$$\Delta_{\perp} = \Delta - \Delta_{\parallel}, \quad \Delta_{\perp} \perp \mathbf{z}. \quad (2)$$

The parallel component $\Delta_{\parallel}$ lies along the reference direction $\mathbf{z}$, scaling its magnitude by α , whereas $\Delta_{\perp}$ lies in the $(d-1)$ -dimensional orthogonal subspace, carrying the direction-changing degrees of freedom.

We apply this decomposition across two primary sites: in *residual space*, Δ is the sub-layer update added to the incoming state $\mathbf{z}$; in *attention value space*, Δ is the pre-projection aggregate relative to the token’s own value vector $\mathbf{z}$.

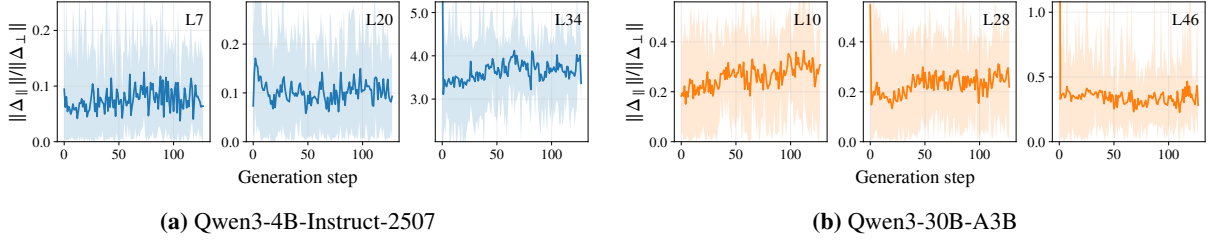


Figure 1: Parallel components persist across depth. For each accumulated attention-plus-MLP update, $\Delta_{t,||}^l$ is its projection along the incoming hidden state, whereas $\Delta_{t,\perp}^l$ is orthogonal to that state. Curves show the ratio $\|\Delta_{t,||}^l\|/\|\Delta_{t,\perp}^l\|$ at three sampled layers per model across generation steps; lines are means and bands span the observed range. Ratios above one indicate parallel dominance.

3.2 Transformer Residual Updates

In a transformer with hidden dimension d , each residual sub-layer adds an update to the current representation:

$$\mathbf{z}^{l+1} = \mathbf{z}^l + \Delta^l. \quad (3)$$

The update Δ^l denotes the output of an individual sub-layer (self-attention Attn or MLP) or the net update of the full transformer block. The reference state $\mathbf{z}_t^l$ is taken as the representation immediately preceding the addition: the sub-layer input for individual modules, or the block input for the accumulated update. We write Δ_t^l for the corresponding residual contribution at token position $t \in \{1, \dots, T\}$.

Using the decomposition in Equation 1, the parallel component of a residual update is:

$$\Delta_{t,||}^l = \alpha_t^l \mathbf{z}_t^l, \quad \alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}, \quad (4)$$

where the scalar α_t^l measures the magnitude of the parallel update relative to the incoming state $\mathbf{z}_t^l$. Substituting this decomposition into the residual addition yields

$$\mathbf{z}_t^{l+1} = (1 + \alpha_t^l) \mathbf{z}_t^l + \Delta_{t,\perp}^l, \quad \Delta_{t,\perp}^l \perp \mathbf{z}_t^l, \quad (5)$$

where the parallel component rescales the incoming representation by $(1 + \alpha_t^l)$, while $\Delta_{t,\perp}^l$ introduces orthogonal direction-changing information.

Figure 1 reports the relative magnitude $\|\Delta_{t,||}^l\|/\|\Delta_{t,\perp}^l\|$ across generation steps at three sampled layers for two Qwen3 models. This ratio quantifies how much of a residual update aligns with the existing representation versus redirects it. The persistence of substantial parallel components across depth motivates investigating when they affect model behavior and in which representation spaces those effects remain robust.

4 Directional Interventions

We instantiate directional decomposition across two complementary spaces: the residual stream (Section 4.1) and the attention value space (Section 4.2). We then unify these sites into a shared component-scaling framework (Section 4.3) to systematically manipulate parallel and perpendicular updates. Figure 2 summarizes both intervention sites and their diagonal view.

4.1 Residual-Level Decomposition

Set $\Delta = \Delta_t^l$ (the residual update at layer l , token position t) and $\mathbf{z} = \mathbf{z}_t^l$ (the pre-update hidden state). For layer and module profiles, we report calibration-corpus averages of the per-token residual parallel ratio:

$$r_t^l = r(\Delta_t^l, \mathbf{z}_t^l) = \frac{|\alpha_t^l| \|\mathbf{z}_t^l\|}{\|\Delta_{t,\perp}^l\|}, \quad (6)$$

$$\alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}.$$

This residual-level view serves as the shared observable for the rest of the paper. The resulting diagnostic is block-agnostic. We apply it to (i) the Attn sub-layer output, (ii) the MLP sub-layer output, and (iii) their accumulated effect at the transformer-block level.

4.2 Value-Space Decomposition

Inside attention, let $\mathbf{x}_s^{\text{attn}}$ denote the input to the value projection, including the model’s pre-attention normalization. Each source token s is mapped to a value vector $\mathbf{v}_s = W_V \mathbf{x}_s^{\text{attn}}$. For query token t , let $\mathcal{A}_{ts}$ denote a linear attention-mixing operator on the value space. This notation does not assume a particular head structure; standard multi-head attention is the block-diagonal

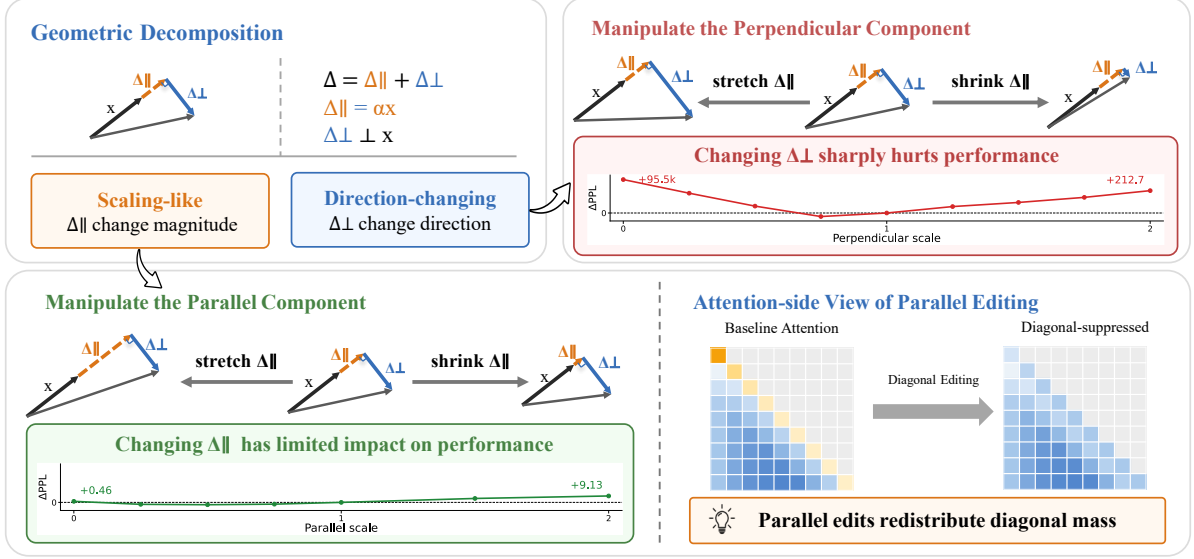


Figure 2: Geometric decomposition and component scaling. Residual updates and attention value aggregates are decomposed into parallel and perpendicular components and scaled independently; parallel-only scaling can be expressed as an attention-diagonal change. Labels report raw ΔPPL relative to the scale-one no-op (lower is better); curves are rescaled only for display.

instance whose head blocks carry their corresponding attention weights. The pre-output-projection aggregate is

$$\mathbf{o}_t = \sum_{s \leq t} \mathcal{A}_{ts} \mathbf{v}_s. \quad (7)$$

Within this space the natural reference is the current token’s own value $\mathbf{v}_t$: the direction $\mathbf{o}_t$ would take if attention routed nothing from other positions.

$$\mathbf{o}_{t,\parallel} = \frac{\mathbf{o}_t \cdot \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{o}_{t,\perp} = \mathbf{o}_t - \mathbf{o}_{t,\parallel}. \quad (8)$$

Residual-space geometry does not isolate this self-value-aligned structure because the output projection mixes the aggregate into the residual stream.

Direct Self-Message Preservation. In naive value-space decomposition (Equation 8), the query token’s self message $\mathbf{d}_t = \mathcal{A}_{tt} \mathbf{v}_t$ is colinear with $\mathbf{v}_t$ and absorbed into $\mathbf{o}_{t,\parallel}$. Naive parallel removal ($s^{(\parallel)} = 0$) thus extinguishes the token’s identity carrier alongside cross-token features. To isolate *contextual* magnitude modulation while preserving self-representation, we decouple the aggregate:

$$\begin{aligned} \mathbf{o}_t &= \underbrace{\mathcal{A}_{tt} \mathbf{v}_t}_{\mathbf{d}_t \text{ (self)}} + \underbrace{\sum_{s < t} \mathcal{A}_{ts} \mathbf{v}_s}_{\mathbf{c}_t \text{ (non-self)}}, \\ \mathbf{c}_{t,\parallel} &= \frac{\mathbf{c}_t \cdot \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{c}_{t,\perp} = \mathbf{c}_t - \mathbf{c}_{t,\parallel}. \end{aligned} \quad (9)$$

We scale only the parallel and perpendicular components of the non-self aggregate and then restore the unchanged self message:

$$\tilde{\mathbf{o}}_t^{\text{excl}} = \mathbf{d}_t + s^{(\parallel)} \mathbf{c}_{t,\parallel} + s^{(\perp)} \mathbf{c}_{t,\perp}. \quad (10)$$

We refer to this operation as *exclude-self* scaling: it neither masks the self-attention edge nor renormalizes the attention row.

Value-space decomposition applies only to attention aggregates. For MLPs, the main analysis decomposes the sub-layer output in residual space; Appendix D separately applies the same projection geometry to the post-gating carrier and grouped down-projected contributions.

4.3 Component Scaling and Diagonal Form

For either attention-side site, write $\mathbf{y}_t = \sum_{s \leq t} \mathcal{M}_{ts} \mathbf{m}_s$ and decompose $\mathbf{y}_t$ relative to its reference $\mathbf{r}_t$. Value-space editing uses $(\mathcal{M}_{ts}, \mathbf{m}_s, \mathbf{r}_t) = (\mathcal{A}_{ts}, \mathbf{v}_s, \mathbf{v}_t)$. Residual-space attention editing uses $\mathcal{B}_{ts} = W_O \mathcal{A}_{ts}$ and $(\mathcal{M}_{ts}, \mathbf{m}_s, \mathbf{r}_t) = (\mathcal{B}_{ts}, \mathbf{v}_s, \mathbf{z}_t^l)$, keeping head-dependent mixing explicit. Component scaling then uses the shared intervention

$$\tilde{\mathbf{y}}_t = s^{(\parallel)} \mathbf{y}_{t,\parallel} + s^{(\perp)} \mathbf{y}_{t,\perp}. \quad (11)$$

The no-op is $s^{(\parallel)} = s^{(\perp)} = 1$, and parallel removal sets $s^{(\parallel)} = 0, s^{(\perp)} = 1$. Exclude-self scaling applies the same form to $\mathbf{c}_t$ and restores $\mathbf{d}_t$ as in Equation 10.

The same edit can also be read on the attention map itself, connecting it to self-directed attention mass (Section 2). Holding all off-diagonal messages fixed, a diagonal update realizes parallel-only scaling when

$$\Delta \mathbf{y}_{tt} = (s^{(\parallel)} - 1) \mathbf{y}_{t,\parallel}. \quad (12)$$

Appendix B gives an operator realization, scalar attention-diagonal forms, removed cases, and the applied-output audit.

5 Experimental Setup

Intervention implementation. All inference-time edits operate strictly during the forward pass and leave model weights untouched. We apply component scaling across decoder layers in two distinct sites: the residual stream and attention value space. In value space, we explicitly evaluate both *full-aggregate* scaling (decomposing the complete attention aggregate $\mathbf{o}_t$ relative to $\mathbf{v}_t$) and *exclude-self* scaling (Equation 10). Because the query token’s direct self message $\mathbf{d}_t = A_{tt}\mathbf{v}_t$ is intrinsically parallel to $\mathbf{v}_t$, naive parallel removal on the full aggregate inadvertently suppresses the token’s primary identity carrier along with cross-token updates, causing severe degradation (Section 6.2). Exclude-self scaling preserves $\mathbf{d}_t$ to isolate whether cross-token contextual magnitude modulation is truly redundant. Unless explicitly marked otherwise, value-space evaluations use exclude-self scaling. Appendices A and B provide complete evaluation and implementation details.

Models and benchmarks. We evaluate across dense and mixture-of-experts architectures from the Qwen3 (Qwen Team, 2025), Llama-3 (Llama Team, 2024), and Gemma-3 (Gemma Team, 2025) families across scales up to 30B parameters. Our benchmarks include language modeling perplexity, seven standard zero-shot commonsense and reasoning tasks, long-context retrieval and reasoning (13-task RULER suite (Hsieh et al., 2024) up to 12k context lengths), and post-training compression under 4-bit AWQ (Lin et al., 2024) and Wanda pruning (Sun et al., 2024) (unstructured, 4:8, and 2:4).

Pretraining setup. For training-time investigations, we pretrain GPT-style models from scratch on OpenWebText (Gokaslan et al., 2019) across scales from 296M to 2.7B parameters, comparing

baseline optimization against residual- and value-space parallel removal; Appendix E details architectural configurations and training hyperparameters.

6 Inference-Time Component Editing

6.1 Directional Component Scaling

Probing directional sensitivity. Section 4 formalized the orthogonal decomposition into parallel magnitude modulation and perpendicular semantic steering across residual and value spaces. We first evaluate the functional roles of these components by measuring model sensitivity to continuous inference-time scaling edits. Holding weights frozen, Figure 3 independently varies $s^{(\parallel)}$ (left, $s^{(\perp)} = 1$) and $s^{(\perp)}$ (right, $s^{(\parallel)} = 1$) across three intervention sites on Qwen3-1.7B: value space (exclude-self), residual attention, and residual MLP.

Directional asymmetry: acute perpendicular fragility. Across all three intervention sites, model behavior exhibits an acute *directional asymmetry*. Perpendicular scaling ($s^{(\perp)} \neq 1$) proves extraordinarily hyper-fragile: even minor deviations from identity immediately destabilize the model with steep perplexity surges, while severe attenuation or complete removal ($s^{(\perp)} = 0$) triggers catastrophic breakdown, escalating by multiple orders of magnitude across all sites (reaching tens of thousands of points in value space and residual attention, and exploding into millions of points in residual MLP). This confirms that orthogonal directions govern delicate, non-interchangeable semantic steering. In stark contrast, parallel scaling ($s^{(\parallel)}$) displays a broad, stable tolerance basin, confirming parallel updates regulate representation magnitude rather than categorical semantic trajectories.

Site asymmetry: value space vs. residual branches. Within the parallel axis, the sweeps reveal a pronounced *site hierarchy*. Value-space manipulation (with the direct self message preserved) stays closest to baseline, remaining virtually flat across $s^{(\parallel)} \in [0, 1]$ (shifting PPL by less than a point, and by at most 1.3 points up to scale 3). In the residual stream, Residual(Attn) is intermediate, whereas Residual(MLP) is most fragile, collapsing rapidly outside the unit interval.

This hierarchy demonstrates that robustness is not an inherent property of “parallelness” in the abstract, but depends decisively on representation space and what is preserved. Residual-space par-

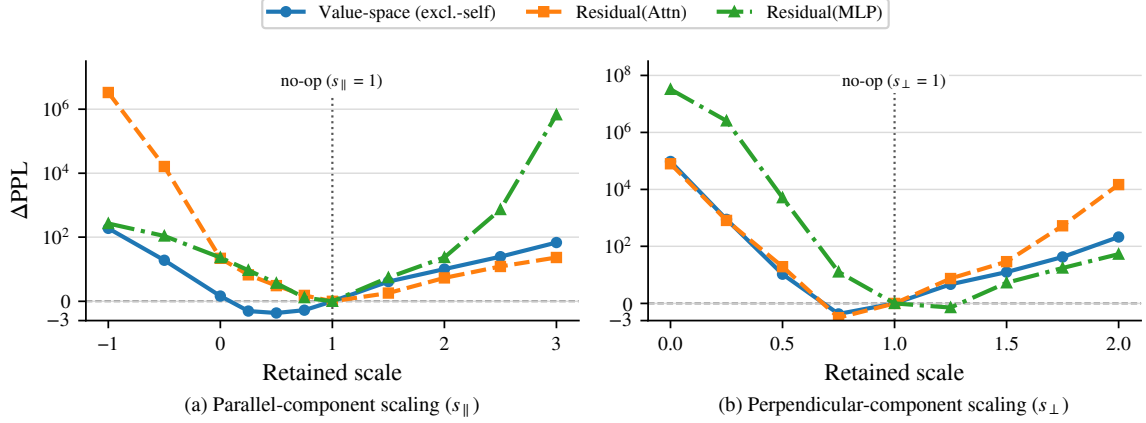


Figure 3: Perplexity under component scaling. The left panel varies $s_{\parallel}$ with $s_{\perp} = 1$; the right varies $s_{\perp}$ with $s_{\parallel} = 1$. Evaluated on Qwen3-1.7B (WikiText-2), ΔPPL is edited minus the matched scale-one no-op, so lower is better. All decoder layers are edited; value-space curves use exclude-self scaling.

allel edits modify the accumulated features of the main stream, where parallel updates carry essential cross-layer computation. In contrast, value-space exclude-self editing modulates only cross-token contextual aggregation while leaving the query token’s primary identity carrier ($\mathbf{d}_t = A_{tt}\mathbf{v}_t$) and perpendicular context intact. A parallel effect appears inside the MLP: decomposing the internal post-gating carrier rather than the final sub-layer output similarly preserves performance under parallel removal (Appendix D).

6.2 Cross-Space Representation Editing

Robustness on general tasks. We evaluate downstream zero-shot robustness across seven standard benchmarks on dense (Qwen3-1.7B) and MoE (Qwen3-30B-A3B) models (Table 1), comparing three primary approaches: (1) self-preserving value-space removal (V-Excl.-self), (2) naive full-aggregate value removal (V-Para Rem.), and (3) residual-space and diagonal controls (Attn Para-Rem. and Diag. Rem.).

When parallel removal is applied naively to the entire value aggregate (V-Para Rem.), average performance drops by 8–10 points across both models. This degradation illustrates why isolating self-attention is essential: naive value-space editing suppresses the token’s own identity carrier $A_{tt}\mathbf{v}_t$ alongside cross-token messages, inadvertently disrupting representation propagation.

In stark contrast, preserving the direct self-message while scaling only cross-token aggregation (V-Excl.-self) leaves performance virtually intact across all seven benchmarks—trailing baseline

by only 1.5 points on 1.7B and virtually matching it on the 30B-A3B MoE model (within 0.1 points), with slight gains on several individual tasks. Meanwhile, residual-space attention removal (Attn Para-Rem.) incurs steady drops across tasks, and hard diagonal removal (Diag. Rem.) severely degrades the models because forcing $A_{tt} = 0$ artificially renormalizes attention weights over off-diagonal positions. Together, these comparisons establish that cross-token parallel updates are functionally redundant, provided direct self-representations are retained.

Context-length scalability. Table 2 reinforces this distinction as context expands (RULER on Llama-3.2-3B from 4k to 12k). Retaining half of the parallel component ($s^{(\parallel)} = 0.5$) with the self-message fixed closely tracks the unedited baseline across all sequence lengths (within 0.2 points at 4k and 2.4 points at 12k), consistently outperforming full-aggregate scaling. Under complete removal ($s^{(\parallel)} = 0$), full-aggregate editing collapses by 26–40 points, whereas exclude-self retains over 75% accuracy at 4k and maintains substantially higher resilience out to 12k. Table 8 in the Appendix details 13-task breakdowns and multi-model evaluations.

Attention diagonal views for parallel editing.

All three attention-side interventions admit closed-form expressions as effective diagonal modifications on the attention matrix (Section 4.3). Figure 4 compares the empirical attention map of Qwen3-4B (layer 19, head 6) under baseline against the residual-space and value-space diagonal views (see

Table 1: Task-level comparison of residual-, value-space, and diagonal controls. Entries are absolute scores (%); higher is better and Avg. is unweighted. Both value-space rows set $s_{\parallel} = 0$. V-Para Rem. edits the full value aggregate; V-Excl.-self preserves $A_{tt}v_t$. Diag. Rem. zeros A_{tt} and renormalizes over non-self positions. Attn Para-Rem. and Diag. Rem. use separate matched baselines. Differences reported in the text are computed before rounding. Bold marks the best baseline/value-space result.

Model	Setting	OBQA	PIQA	Wino	ARC-C	HSwag	BoolQ	RTE	Avg.
Qwen3-1.7B	Baseline	37.20	72.47	60.69	53.07	60.16	77.65	70.04	61.61
	Attn Para-Rem.	38.20	72.31	56.59	50.09	58.74	71.38	66.79	59.16
	V-Para Rem.	34.40	69.48	54.85	41.38	54.04	66.27	53.43	53.41
	V-Excl.-self	39.20	71.76	60.54	49.15	59.88	74.46	65.70	60.10
	Diag. Rem.	26.20	51.36	49.17	24.49	26.66	40.70	55.60	39.17
Qwen3-30B-A3B	Baseline	44.80	80.69	70.24	69.97	77.79	88.53	81.95	73.42
	Attn Para-Rem.	44.00	78.73	66.85	65.10	73.87	88.04	78.34	70.70
	V-Para Rem.	41.20	76.22	64.56	59.04	69.57	70.89	62.45	63.42
	V-Excl.-self	44.60	79.87	72.45	68.34	78.13	86.36	83.39	73.31
	Diag. Rem.	30.60	57.67	48.46	26.96	34.78	38.59	47.65	40.68

Table 2: Long-context performance across sequence lengths. Reported scores are mean accuracy (%) across all 13 RULER tasks on Llama-3.2-3B for context lengths from 4k to 12k tokens. V-Full scales the entire value aggregate; V-Excl.-self preserves direct self messages d_t and scales only cross-token context.

Method	$s_{\parallel}$	4k	8k	12k
Baseline	1.0	86.91	82.09	79.30
V-Full	0.5	84.81	78.80	74.76
V-Excl.-self	0.5	86.69	80.80	76.88
V-Full	0.0	60.99	48.61	39.17
V-Excl.-self	0.0	75.64	67.97	62.02

Table 5 in the Appendix for complete mathematical derivations).

The residual-space diagonal view induces volatile positive and negative adjustments (spanning -1.00 to 0.76), whereas exclude-self value-space editing preserves direct self-token routing with bounded, non-positive diagonal shifts. Crucially, this localized stability does not mean value-space editing is gentler: auditing the post- W_O attention output on Qwen3-0.6B reveals that removing parallel context across non-self tokens induces a much larger perturbation norm than residual-space removal (49.10% versus 25.15% of unedited branch norm; Table 6). That value-space editing preserves model capabilities despite altering attention outputs nearly twice as much proves that stability is governed by preserving self-identity routing and residual propagation trajectories, rather than minimizing isotropic perturbation magnitude. Appendix C details head-level localization.

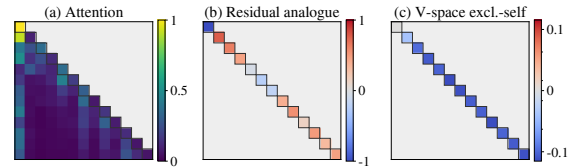


Figure 4: Attention diagonal views of parallel editing. For full parallel removal ($s_{\parallel} = 0$), the left panel shows empirical attention weights (Qwen3-4B, layer 19, head 6); the middle shows the residual-space attention-map view, and the right shows the exclude-self value-space change. The latter two color scales encode ΔA_{tt} ; off-diagonal entries are unchanged.

6.3 Compression Diagnostics

The removal experiments establish that model capabilities are exceptionally sensitive to direction-changing components of activation updates. We next ask how this geometric principle manifests under post-training compression. Compression alters model parameters, inducing an error between uncompressed and compressed updates rather than an explicit activation edit. Decomposing this error into parallel and perpendicular components provides a principled test: if orthogonal steering is the behaviorally critical component, superior compression algorithms should systematically exhibit lower perpendicular distortion, whereas parallel error should be far less diagnostic.

Concretely, at each layer, let Δ_{base} denote the uncompressed baseline update and Δ_{comp} denote the compressed update. The local compression error is $e = \Delta_{\text{comp}} - \Delta_{\text{base}}$. We decompose e relative to the baseline update direction: $e_{\parallel} = \text{proj}_{\Delta_{\text{base}}} e$ and

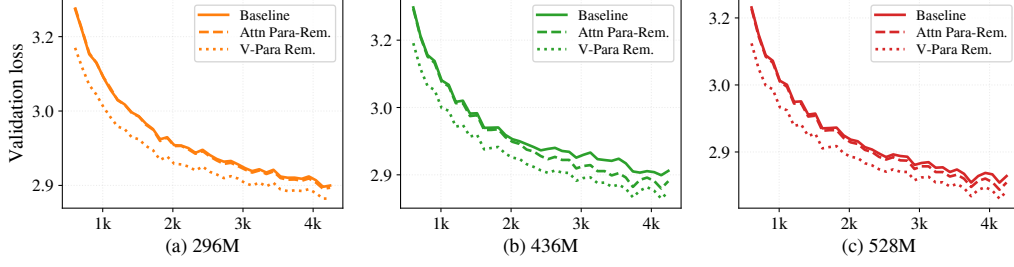


Figure 5: Training-time parallel removal lowers validation loss. The displayed OpenWebText (Gokaslan et al., 2019) loss trajectories show baseline, Attn Para-Rem., and V-Para Rem. for matched configurations across 296M, 436M, and 528M model sizes. Both interventions set $s^{(\parallel)} = 0$ and retain the perpendicular component.

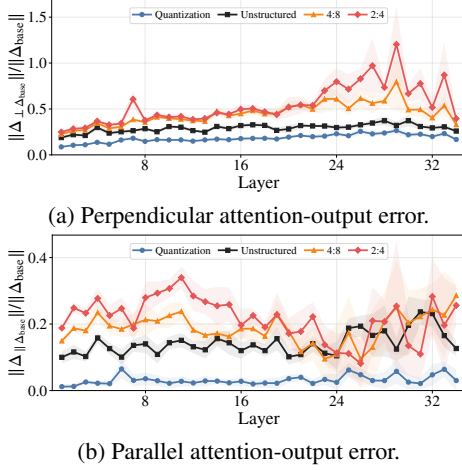


Figure 6: Directional decomposition of compression error. Curves are layerwise means, with bands showing one standard deviation over 12 fixed prompts, comparing compressed and baseline Qwen3-4B attention updates. The upper and lower panels show perpendicular and parallel error, each normalized by the full baseline-update norm (higher is more distortion). Settings are 4-bit AWQ and 50% Wanda pruning with unstructured, 4:8, or 2:4 masks.

$e_{\perp} = e - e_{\parallel}$. Figure 6 reports the normalized component norms $\|e_{\perp}\|/\|\Delta_{\text{base}}\|$ and $\|e_{\parallel}\|/\|\Delta_{\text{base}}\|$ across layers of Qwen3-4B under 4-bit AWQ and 50% Wanda pruning (unstructured, 4:8, and 2:4).

The results strongly corroborate this directional hypothesis: perpendicular error curves cleanly separate the four compression regimes in strict alignment with their downstream fidelity (4-bit AWQ achieves the lowest error, followed by unstructured, 4:8, and 2:4 Wanda). In contrast, parallel error curves heavily interleave and fail to provide a consistent ranking. This clarifies why conventional isotropic L_2 error can fail to predict degradation: isotropic distance conflates benign magnitude shifts with disruptive semantic rotations. Appendix F substantiates this mechanism across the MLP branch and full block updates, showing that perpendicular distortion dominates total compression error

($r > 0.97$, Table 12) and connecting this geometry to cosine-based pruning diagnostics.

7 Training-Time Allocation

In standard Transformer pretraining, attention updates conflate colinear magnitude rescaling with orthogonal contextual steering. *We investigate whether suppressing parallel attention updates throughout training shifts how optimization allocates representational capacity across direction-changing components.* We train GPT-style language models from scratch on standard next-token prediction, enforcing parallel removal during both optimization and evaluation to eliminate train-evaluation mismatch. We evaluate these interventions through validation trajectories and downstream benchmarks; Appendix E gives model configurations and gated-scaling controls.

Figure 5 plots OpenWebText (Gokaslan et al., 2019) validation loss trajectories across model sizes (296M, 436M, and 528M). Suppressing parallel attention updates shifts the learning trajectory downward from early training stages, and this advantage persists consistently through the final checkpoint rather than emerging only as an endpoint artifact.

Table 3 reports downstream zero-shot accuracy across six standard benchmarks for the scaled 1.4B and 2.7B models. Both interventions improve downstream performance over the baseline at both scales, with value-space parallel removal achieving the largest gains (+0.7 points on 1.4B and +1.5 points on 2.7B). Together with training trajectories, these findings demonstrate that directional decomposition provides an effective inductive bias: suppressing parallel updates directs attention capacity toward orthogonal contextual steering, improving downstream generalization.

Appendix E details task-level scores for both larger models and the gated-scaling control, where fixed removal maintains the lowest training loss.

Table 3: Downstream generalization across model scales. Unweighted mean accuracy over six benchmarks (ARC-Easy, BoolQ, HellaSwag, OpenBookQA, PIQA, WinoGrande); ΔAvg is relative to matched baseline. Attn and V-Para denote residual and value parallel removal.

Model	Setting	Avg	ΔAvg
1.4B	Baseline	58.5	0.0
	Attn Para-Rem.	58.8	+0.3
	V-Para Rem.	59.2	+0.7
2.7B	Baseline	60.2	0.0
	Attn Para-Rem.	60.9	+0.7
	V-Para Rem.	61.7	+1.5

8 Discussion

Geometric Duality of Representation Evolution.

Our decomposition reveals that Transformer layer updates execute two distinct geometric functions: *magnitude modulation* ($\Delta_{\parallel}$) and *semantic steering* ($\Delta_{\perp}$). Perpendicular updates rotate representations into orthogonal subspaces housing factual and syntactic distinctions; because these coordinates are finely calibrated, $\Delta_{\perp}$ is acutely fragile. Conversely, parallel updates act as adaptive gain controllers, amplifying existing trajectories without altering directional meaning. The extensive tolerance basin of $\Delta_{\parallel}$ confirms that Transformers possess inherent resilience to magnitude variations, preserving semantic intent despite substantial scale disruption.

Beyond Isotropic Compression Error. Quantization and pruning minimize isotropic L_2 error $\min \|\Delta - \hat{\Delta}\|_2^2$, assuming uniform sensitivity across $\mathbb{R}^d$. Our directional framework exposes this flaw: update-aligned errors ($\Delta_{\parallel}^{\text{err}}$) fall within the benign tolerance basin, whereas orthogonal deviations ($\Delta_{\perp}^{\text{err}}$) directly distort semantic trajectories. This explains why compression methods with comparable global L_2 error diverge in downstream fidelity. Measuring directional distortion establishes $\Delta_{\perp}^{\text{err}}$ as a causal predictor of degradation, showing compression objectives should penalize directional rotation over uniform distance.

Architectural Implications and Pretraining Dynamics.

In standard self-attention, token mixing is structurally entangled with self-representation amplification, allocating parameter capacity to redundant parallel scaling. Suppressing parallel updates relieves attention from these duties, guiding optimization to dedicate capacity toward directional contextual steering. This inductive bias ac-

celerates convergence, lowering validation-loss trajectories and improving downstream generalization across model scales from 296M to 2.7B parameters. These findings provide principled geometric support for emerging architectures that decouple contextual routing from magnitude modulation (such as gated attention and query-key normalization).

9 Conclusion

We formalize representation evolution in Transformers through orthogonal decomposition, separating updates into parallel magnitude modulation ($\Delta_{\parallel}$) and perpendicular semantic steering ($\Delta_{\perp}$). Across diverse pretrained language models, interventions reveal a universal directional asymmetry: representations tolerate substantial colinear scaling along their trajectory, yet exhibit acute fragility under perpendicular perturbations. Furthermore, this stability is strongly site-dependent: value-space aggregation proves markedly more robust than residual-space edits by preserving direct token routing.

This geometric perspective resolves key challenges across the model lifecycle. In compression, directional error decomposition clarifies why isotropic L_2 distance often fails to predict behavioral loss: perpendicular distortion causally drives degradation, whereas parallel shifts are largely benign. In pretraining, suppressing parallel attention updates acts as an effective inductive bias, directing capacity toward orthogonal contextual routing. Across scales up to 2.7B parameters, this constraint lowers validation loss and improves downstream performance, establishing directional geometry as a foundation for representation analysis.

Limitations

Our experiments cover decoder-only language models across multiple scales, language tasks, inference-time interventions, compression settings, and from-scratch training configurations. Although the same geometric picture is consistent across the evaluated settings, its behavior under other architectures, modalities, and post-training regimes remains to be established. Future work should test whether these directional patterns persist across such settings and connect layer-level geometry more directly to downstream behavior. We use the proposed edits primarily as analysis tools; turning them into practical training or inference methods requires further validation.

Ethical Considerations

Directional decomposition provides a concrete tool for auditing representation change and screening compression or interventions before downstream evaluation. This screening requires no retraining and is designed to complement task-specific testing and human review. Responsible deployment should evaluate prompt- and context-specific behavior as well as bias, privacy, security, and high-stakes use. Access controls and downstream monitoring are appropriate when deploying editing capabilities.

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A Evaluation Details

Evaluated models. The editing and diagnostic experiments use Qwen3 (Qwen Team, 2025) base checkpoints, except Figure 1, which uses Qwen3-4B-Instruct-2507. Long-context evaluation uses Llama-3.2-3B (Llama Team, 2024).

Benchmark evaluation. Tasks are evaluated via Language Model Evaluation Harness (Gao et al., 2024) and RULER (Hsieh et al., 2024), following standard configurations (Wang et al., 2018; Hendrycks et al., 2021; Cobbe et al., 2021; Reddy et al., 2019; Dua et al., 2019; Lin et al., 2022; Clark et al., 2018; Zellers et al., 2019). Table 4 gives the default few-shot setting and metric for each task.

Table 4: Default evaluation task settings. Acc. denotes accuracy; Norm denotes length normalization.

Task	Shots	Metric	Task	Shots	Metric
ARC-Chal.	25	Acc. (Norm)	MMLU	5	Acc.
ARC-Easy	0	Acc. (Norm)	OpenBookQA	0	Acc. (Norm)
BoolQ	0	Acc.	PIQA	0	Acc. (Norm)
CoQA	0	F1	RTE	0	Acc.
DROP	0	F1	TruthfulQA-MC	0	Acc.
GSM8K	5	Exact match	TruthfulQA gen.	0	Rouge-L
HellaSwag	10	Acc. (Norm)	WinoGrande	5	Acc.

B Intervention Details

Attention-diagonal editing. Equation 12 defines the requested change in the parallel component. Let $\mathbf{m}_t$ be the self message, $\mathbf{r}_t$ the reference, and $\mathcal{B}_{tt}$ its linear map into the edited space. One minimum-Frobenius-norm realization of $\Delta \mathbf{y}_{tt} = \Delta \mathcal{B}_{tt} \mathbf{m}_t$ is $\Delta \mathcal{B}_{tt}^* = (s^{(\parallel)} - 1) \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\|\mathbf{m}_t\|^2 \|\mathbf{r}_t\|^2} \mathbf{r}_t \mathbf{m}_t^\top$. For a structured operator family, the same construction uses the Frobenius projection of $\mathbf{r}_t \mathbf{m}_t^\top$ onto the admissible operator subspace. Table 5 gives the resulting scalar forms. For the exclude-self variant in our primary value-space results, Eq. 10 applies projection to $\sum_{s < t} A_{ts} \mathbf{v}_s$, restoring direct $A_{tt} \mathbf{v}_t$.

Applied strength of the two edits. Section 6.2 and Table 5 show that exclude-self value-space editing preserves the self contribution $A_{tt} \mathbf{v}_t$, inducing bounded non-positive shifts along the attention diagonal, whereas residual-space removal forces volatile signed swings (-1.00 to $+0.76$). However, diagonal stability does not imply a gentler intervention. Table 6 audits both Full No-Para interventions at the post- W_O output, comparing edited $\tilde{\mathbf{y}}_t$ against baseline $\mathbf{y}_t$. The exclude-self value-space edit induces a substantially larger perturbation on every metric: 49.10% norm change and 25.89% energy

Table 5: Attention-diagonal forms for component scaling. Value-space form is exact; residual form matches the scalar constraint; “removed case” sets $s^{(\parallel)} = 0$ while retaining the perpendicular component.

Quantity	Form
<i>Full-aggregate value space</i>	
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{\mathbf{v}_t^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}$.
Removed case	$\tilde{A}_{tt} = - \sum_{s < t} A_{ts} \frac{\mathbf{v}_s^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}$.
<i>Residual space</i>	
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{(\Delta_t^{\text{Attn}})^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}$.
Removed case	$\tilde{A}_{tt} = A_{tt} - \sum_{s \leq t} A_{ts} \frac{(W_O \mathbf{v}_s)^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}$.

removal, versus 25.15% and 9.32% for residual-space removal. This divergence reveals a geometric asymmetry: residual-space removal prunes only the 1D projection along incoming $\mathbf{z}_t$, which is a smaller fraction of branch norm yet fatally disrupts inter-layer residual propagation. Conversely, value-space editing zeros parallel components across the non-self aggregate $\sum_{s < t} A_{ts} \mathbf{v}_s$, discarding significant contextual energy while preserving the primary diagonal identity channel. Preserving capabilities despite perturbing outputs nearly twice as heavily shows that stability depends on structural alignment rather than isotropic magnitude.

Table 6: Applied strength of attention Full No-Para interventions. Full No-Para sets $s_{\parallel} = 0$ on Qwen3-0.6B. Perturbation norm is $\|\tilde{\mathbf{y}} - \mathbf{y}\|/\|\mathbf{y}\|$, retained norm is $\|\tilde{\mathbf{y}}\|/\|\mathbf{y}\|$, and perturbation energy is $\|\tilde{\mathbf{y}} - \mathbf{y}\|^2/\|\mathbf{y}\|^2$ relative to baseline $\mathbf{y}$.

Geometry	Perturb. norm	Perturb. energy	Retained norm
Attention residual-space	25.15%	9.323%	95.01%
Attention exclude-self value-space	49.10%	25.89%	83.36%

Computational cost. We apply Eq. 11 through forward hooks replacing only the selected tensor, leaving subsequent computation unchanged. The fused kernel combines dot products, norms, and projection updates with FlashAttention-2 (Dao, 2024); Table 7 reports wall-clock overhead.

Table 7: Measured pre- W_O projection overhead. Wall-clock overhead relative to unedited baselines on an RTX 6000 Ada (BF16; 2,048 prefill, 512-token decode with CUDA Graph replay; 0.7B training with batch 1, length 2,048).

Workload	Qwen3-0.6B	Qwen3-4B	Qwen3-14B
Prefill	2.47%	1.25%	1.11%
Static graph decode	0.83%	0.36%	0.25%
Training (0.7B)	3.39%		

C Additional Editing Results

Zero-shot parallel editing. To test whether value-space parallel tolerance is an artifact of Qwen3 or short contexts, Table 8 evaluates retained-scale editing ($s_{\parallel} = 0.5$) under long-context retrieval on RULER-4k across three distinct model families. Across Gemma-3-12B, Llama-3.2-3B, and Qwen3-4B/8B, scores remain within 0.5 points of matched unedited baselines, confirming that value-space parallel tolerance generalizes across model families and long-context regimes.

Table 8: Cross-model retained-scale RULER-4k evaluation. V-Excl.-self preserves direct self contribution and sets $s_{\parallel} = 0.5$ for the non-self aggregate; Δ is edited minus baseline (computed before rounding).

Model	Baseline	V-Excl.-self, $s_{\parallel}=0.5$	Δ
Gemma-3-1B-IT	68.0	62.8	-5.3
Gemma-3-12B-IT	94.0	94.1	+0.1
Llama-3.2-3B	86.9	86.7	-0.2
Qwen3-0.6B	79.8	75.1	-4.8
Qwen3-1.7B	88.1	84.9	-3.2
Qwen3-4B	92.6	92.4	-0.2
Qwen3-8B	94.2	93.7	-0.5

Head-level localization. Our primary interventions suppress parallel components across all attention heads jointly. A key mechanistic question is whether sensitivity originates from isolated outlier heads or reflects distributed capacity across the multi-head ensemble, and whether single-head edits already produce disruption. To test this, we compare isolated single-head parallel removal against joint all-head removal across four Qwen3 scales (Figure 7). This reveals two core properties. First, *single-head resilience*: editing any single head produces negligible loss change (blue envelopes remain near zero), showing mutual redundancy where neighboring heads compensate for isolated subspace removal. Second, *super-linear joint degradation*: simultaneously suppressing parallel components across all heads (orange curve) degrades loss by orders of magnitude beyond the single-head envelope; multi-head edits eliminate the layer’s collective representational channel. Finally, layer 0 exhibits the sharpest joint disruption across scales, showing early token routing relies critically on shared multi-head parallel coordinates.

D MLP-Internal Geometry

Residual-space editing observes only final MLP output $y = W_{\text{down}}h \in \mathbb{R}^d$. Inside the MLP, however, the post-gating carrier $h = \phi(W_{\text{gate}}x) \odot W_{\text{up}}x \in \mathbb{R}^{d_{\text{mlp}}}$ has a wider dimension than the

residual state $x \in \mathbb{R}^d$ ($d_{\text{mlp}} > d$), so direct projection between h and x is undefined. To align dimensions while preserving the unedited forward pass, we partition the d_{mlp} coordinates into $G = d_{\text{mlp}}/d$ disjoint blocks $I_g = \{(g-1)d+1, \dots, gd\}$ of width d . By block matrix multiplication, $y = W_{\text{down}}h = \sum_{g=1}^G y_g$, where each carrier slice $h_g = h[I_g] \in \mathbb{R}^d$ and block contribution $y_g = W_{\text{down}}[:, I_g]h_g \in \mathbb{R}^d$ shares dimension d with x .

With directional scaling $\mathcal{P}(u; r) = s_{\parallel} \frac{u \cdot r}{\|r\|^2} r + s_{\perp} \left(u - \frac{u \cdot r}{\|r\|^2} r \right)$, the two internal editing sites in Table 9 are:

$$\tilde{y}_{\text{carrier}} = \sum_g W_{\text{down}}[:, I_g] \mathcal{P}(h_g; x), \quad (13)$$

$$\tilde{y}_{\text{contrib}} = \sum_g \mathcal{P}(y_g; h_g). \quad (14)$$

Site 1 (*grouped h rel. to x*) edits intermediate carrier h_g before down projection; Site 2 (*grouped y_g rel. to h_g*) edits projected contribution y_g before group summation. These controls isolate directional asymmetry within intermediate representations from down-projection effects. On Qwen3-0.6B, full parallel removal shifts fixed C4 loss by at most 0.0301, whereas perpendicular removal increases it by > 13 points (Table 9).

Table 9: MLP-internal component removal. C4 Δ loss (lower is better). Full No-Para/No-Perp sets retained scale to zero; no-op matches baseline.

Geometry	Full No-Para	Full No-Perp
Residual MLP (y relative to x)	+0.1165	+15.9567
Grouped h relative to x	+0.0301	+13.4184
Grouped y_g relative to h_g	+0.0037	+13.3324

Downstream task validation. While fixed C4 loss (Table 9) confirms that internal parallel removal preserves perplexity, cross-entropy averages over easy tokens and can mask structured reasoning degradation. To test functional stability, Table 10 audits Full No-Para ($s_{\parallel} = 0$) and Full No-Perp ($s_{\perp} = 0$) across Qwen3-0.6B/1.7B on Core-MC (9 tasks), MMLU (5-shot), GSM8K, TruthfulQA, and DROP/CoQA. Full No-Para preserves downstream reasoning across scales: Core-MC (50.98 vs. 50.94 on 0.6B; 55.28 vs. 55.19 on 1.7B) and MMLU (52.66 vs. 52.40; 60.39 vs. 60.21) match baselines. Conversely, Full No-Perp collapses catastrophically: MMLU plummets by 25.5/35.8 points to near-chance guessing, and Core-MC falls 14–18 points. Thus, parallel components carry dispensable scaling, while perpendicular updates encode essential factual and reasoning knowledge.

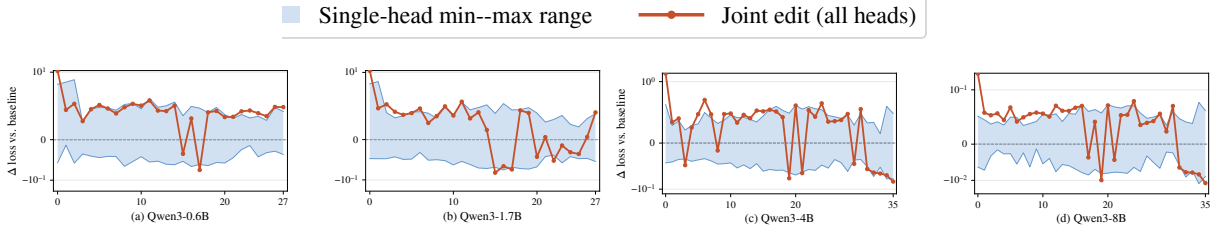


Figure 7: All-layer head sensitivity across Qwen3 scales. At each layer, orange shows joint full removal from all heads; the blue band spans minimum and maximum isolated single-head effects (not a confidence interval). Δ loss is edited minus baseline over 128 fixed C4 texts (positive is worse, symmetric-log axes).

Table 10: Downstream controls for MLP-internal geometry. Core-MC (9 tasks); MMLU (5-shot); GSM8K-S/F (strict/flex EM); TQA (Rouge-L); DROP/CoQA (F1). Collapsed runs marked ‘—’.

Model	Setting	Core-MC	MMLU	GSM-S	GSM-F	TQA	DROP	CoQA
Qwen3-0.6B	Baseline	50.94	52.40	49.43	50.42	18.98	7.76	74.24
	Full No-Para	50.98	52.66	49.96	50.72	15.61	7.66	74.04
	Full No-Perp	36.62	26.89	—	—	—	—	—
Qwen3-1.7B	Baseline	55.19	60.21	68.76	68.92	48.34	7.32	77.06
	Full No-Para	55.28	60.39	68.08	68.01	48.32	7.44	77.69
	Full No-Perp	37.35	24.40	—	—	—	—	—

E Pretraining Setup and Evaluation

Architecture and pretraining setup. Pretraining trajectories (Figure 5) train standard Transformer models from scratch on OpenWebText (Gokaslan et al., 2019) (296M/436M/528M) and FineWeb100BT (Penedo et al., 2024) (1.4B/2.7B) with all-layer value-space parallel removal. Dense configurations use standard GQA: 1.4B ($d = 2048$, 24 layers, 16/4 heads); 2.7B ($d = 2560$, 32 layers, 32/8 heads). Models train on ≈ 104.9 B tokens over 200K steps (batch 256, length 2,048; lr $4 \times 10^{-4}/3 \times 10^{-4}$, 2K warmup, cosine decay, AdamW $\beta = (0.9, 0.95)$, decay 0.1, clip 1.0; $8 \times \text{H100}$ in BF16).

Downstream evaluation results. Table 11 resolves summary downstream averages from Table 3 (§7) into benchmark scores across ARC-E, BoolQ, HSwag, OBQA, PIQA, and WinoGr. Value-space removal consistently outperforms the baseline (+0.7 on 1.4B, +1.5 on 2.7B), confirming robust inductive bias across tasks.

Gated parallel-scaling diagnostic. To test whether parallel scaling requires learned modulation, we compare fixed removal ($s_{\parallel} = 0$) against tokenwise learned gating ($s_{\parallel} = \sigma(w^T x)$) on 1.4B pretraining (Figure 8). Fixed removal maintains lowest loss throughout, showing parallel suppression acts as an inductive bias.

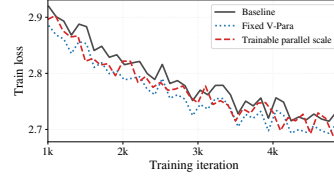


Figure 8: Parallel-scale diagnostic on 1.4B pretraining. FineWeb100BT validation loss vs. step: fixed removal ($s_{\parallel} = 0$) vs. learned gating.

F Extended Compression Geometry

While Figure 6 (§6.3) evaluated directional error at attention outputs (Δ_{attn}), practical compression alters matrices throughout the Transformer block. Figure 9 extends this breakdown to MLP output (Δ_{mlp}) and combined block update ($\Delta_{\text{block}} = \Delta_{\text{attn}} + \Delta_{\text{mlp}}$). Across both sublayers, perpendicular error strictly tracks degradation ($\text{AWQ} < \text{unstructured} < 4:8 < 2:4$), while parallel error is non-monotonic. Table 12 confirms perpendicular error aligns with total distortion ($r \geq 0.97$, $\rho \geq 0.90$), accounting for 87.5%–98.0% of error across layers.

Table 12: Attention-output error alignment on Qwen3-4B.

Setting	$r(e, e_{\perp})$	$\rho(e, e_{\perp})$	$\rho(e, e_{\parallel})$	Perp./Total	Para./Total
Quantization	0.998	0.990	0.596	0.980	0.168
Unstructured	0.970	0.909	0.810	0.891	0.439
4:8	0.986	0.986	-0.135	0.887	0.415
2:4	0.991	0.979	-0.432	0.875	0.425

Alignment with cosine diagnostics. Our directional decomposition mathematically aligns with cosine diagnostics for layer dropping (He et al., 2026). For residual state x and combined block update $\Delta = \alpha x + \Delta_{\perp}$ ($\|\Delta_{\perp}\| = \beta\|x\|$), cosine similarity is $\cos(x, x + \Delta) = (1 + \alpha)/\sqrt{(1 + \alpha)^2 + \beta^2}$. As baseline updates are small ($\|\Delta\|/\|x\| \ll 1$; Figure 10), $\alpha, \beta \ll 1$, yielding $1 - \cos(x, x + \Delta) \approx 0.5(\|\Delta_{\perp}\|/\|x + \Delta_{\parallel}\|)^2$. Angular diagnostics thus evaluate perpendicular steering normalized by parallel magnitude.

Table 11: Task-wise downstream scores for pretrained models. Resolving the summary downstream averages reported in Table 3 (§7) into individual benchmark scores. Avg. is unweighted and ΔAvg is relative to the baseline at the same model scale. Base, Attn, and V-Para denote baseline, residual-space attention parallel removal, and full-aggregate value-space parallel removal.

Model	Setting	ARC-E	BoolQ	HSwag	OBQA	PIQA	WinoGr	Avg	ΔAvg
1.4B	Base	56.0	65.1	60.5	34.7	75.8	58.7	58.5	0.0
	Attn	57.1	63.9	61.3	35.3	76.0	59.1	58.8	+0.3
	V-Para	58.3	62.8	62.1	35.9	76.3	59.8	59.2	+0.7
2.7B	Base	58.4	61.2	66.0	37.1	76.5	61.8	60.2	0.0
	Attn	59.3	62.8	66.7	37.7	77.0	62.1	60.9	+0.7
	V-Para	60.4	64.4	67.1	38.2	77.6	62.6	61.7	+1.5

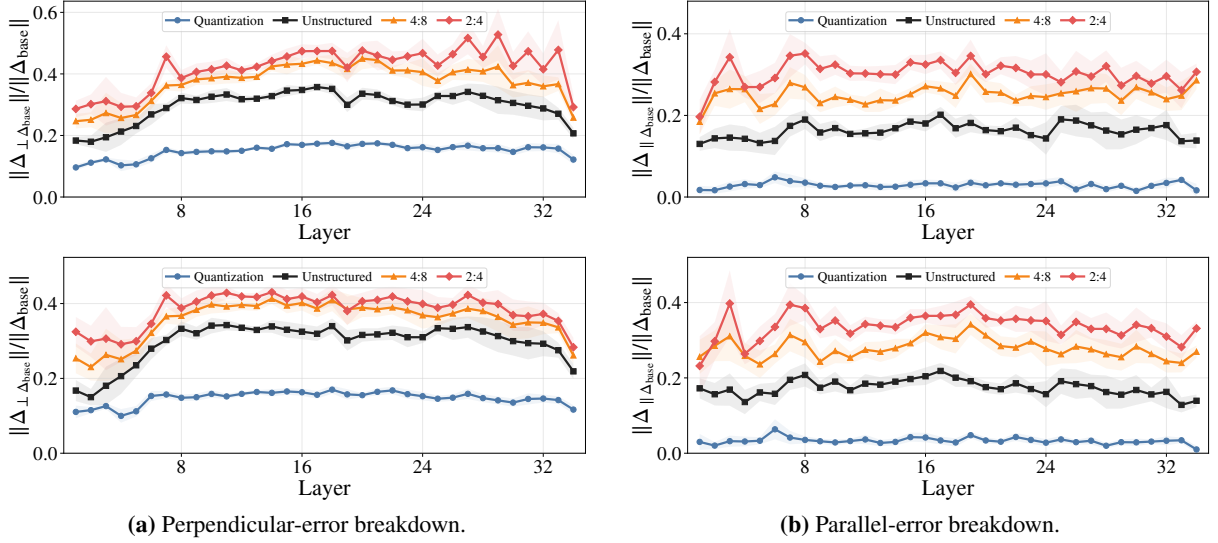


Figure 9: Compression-geometry breakdown across branches. The left and right groups show perpendicular and parallel error, respectively. Within each group, the upper and lower panels show compressed-minus-baseline error for the combined Transformer block update ($\Delta_{\text{block}} = \Delta_{\text{attn}} + \Delta_{\text{mlp}}$) and the MLP output (Δ_{mlp}), each normalized by the corresponding full baseline-update norm. Curves are layerwise means, with bands showing one standard deviation over 12 fixed prompts on Qwen3-4B. Settings are 4-bit AWQ and 50% Wanda pruning with unstructured, 4:8, or 2:4 masks.

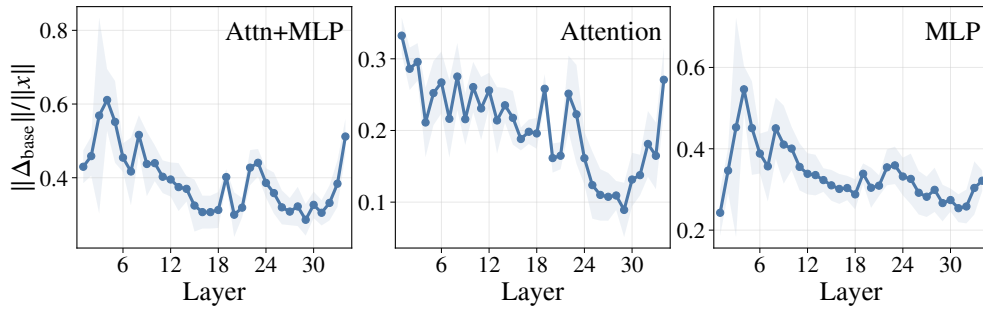


Figure 10: Baseline layer updates are small relative to the residual stream. Means and one-standard-deviation bands for $\|\Delta_{\text{base}}\|/\|x\|$ across 12 prompts for Qwen3-4B layers 1–34, shown for the combined Transformer block update ($\Delta_{\text{block}} = \Delta_{\text{attn}} + \Delta_{\text{mlp}}$), attention output, and MLP output. Values below one indicate updates smaller than the incoming residual stream.